

GE M01 Language Fundamentals and Tools

Personal Note: sections where yellow fill has been replaced with green fill notes a section is completed.

Points: 25 (See Rubric in Canvas)

Due Date: Due date listed in Canvas but some sections will be due as class participation before the final due date. Create Calendar reminders

- Final submission will be accepted up to 24 hours after the due date with a 10% penalty. Meaning if you turn it in at 12:01 am of the next day you will be deducted 10% of the total points from your score.
- If the assignment is more than 24 hours late, it will be a 0.

Submission: Upload files separately and do not upload a zip file.

- **Two Document Files as PDF or Word document:** This document with your answers in the highlighted areas unless otherwise stated and Technical Document
- **Two Java Files (. java files NOT CLASS FILES):** Code from
 - [2.1 Explore GuessNumber Code](#)
 - [2.2 Calculate Grades](#)

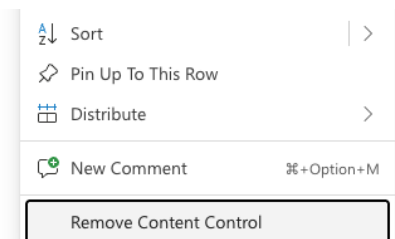
Objectives:

- Analyze and apply module 01 concepts
- Develop effective technical documentation to demonstrate your understanding of technical concepts, with a focus on clarity and professional presentation.
- Apply version control and debugging practices using Eclipse and GitHub.

Effort: You are encouraged to collaborate to discuss concepts and explore writing code together. Remember

- Review lectures and links in the lectures to information first.
- Follow [CS Academic Integrity and AI Policy - Harding](#)
- If you find you don't understand the information in the lecture reach out to **Deb, Elaine or Heriberto**

If you download this as a word document and you have any problems typing your answer in the tables you must click on 3 dots and remove content control.



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Part 0 Learning and Collaborating Reflection

The reality of computer science is that there are so many languages, technologies and methodologies available and it is constantly evolving with new ones. So the goal is not just to understand the current technologies but developing skills to learn any new technology.

0.1 Working on Teams

In this course, we'll work together on in-class activities, labs, and discussions. Good collaboration isn't just about dividing work — it's about building understanding together. Read [10 Collaboration Skills Examples](#) and answer the following

1. Pick two of the 10 skills that you think are strengths you bring to a team and why.

Two of the 10 skills I see as strengths I can bring to a team include **active listening and receiving feedback**.

My current research position has provided me the opportunity to develop and strengthen these two skills. What I have seen myself develop in these skills through this experience is being able to use active listening to track what content is most beneficial and important when taking notes for a group, and then being able to use this active listening to receive and implement feedback. This active listening skill has also shown helpful in how I receive feedback. I am able to better understand the key action items and focus points of a piece of feedback. This skill has also helped me feel confident in following up on feedback to ensure I fully understood and applied someone's advice to the intended item or task.

I also think this experience has provided me a lot of practice in giving academic feedback as well, although I do still feel I have a long way to go in feeling fully confident in this skill. I really appreciate feedback and feel like I

do very well at implementing feedback from others into my work, but I definitely find myself still nervous and unsure about giving feedback to others, especially when they are much further in a program than I am or they have been working on a project longer.

2 Pick two of the 10 skills you think are most difficult for you and something you can do to try and improve in that area this semester.

Two skills that are most difficult for me include **conflict resolution**, and **giving constructive feedback**.

When I am in spots with folks who have more experience in a subject, or have been on a project longer, it can be hard for me to branch out or challenge a way of thinking or approach to a topic. I often feel like I need to spend much more time reviewing previous journals or publications to feel confident enough to ask questions or give suggestions.

I am definitely still working on feeling comfortable asking questions or brainstorming different problem solving strategies with folks even if I don't feel 110% confident in the background of a project or what the solution should be for a project. I know the most impactful learning experiences come with being upfront about what you do and don't know so that a team or mentor can collaborate with you effectively, and so I have been keeping this in mind frequently to help get over the nerves that come with asserting myself in my education.

I am hopeful that collaborating in groups often will help me build my confidence in a team. I think working with different teams will help me practice using active listening to take notes of group strengths and individual strengths alongside sections members of a group may struggle in so that we can all help each other thrive in this class.

I also hope being able to gain multiple perspectives of how my classmates approach a project will help me to learn how to tailor my feedback to each team-member's learning style so that it can be as beneficial and supportive as possible.

0.2 Learning How to Learn

Learning how to learn is an important component of this course as your career will be full of learning something new. One of my favorite websites is [Train Ugly - How To Get More Out Of Your Practice](#).

1 Listen to this 30 minute podcast [Desirable Difficulties - The Learner Lab](#) and answer the following.

Explain desirable difficulties and how you think it relates to the learning in this class and in your future career.

Desirable difficulties are described by the podcast hosts as a form of learning that implements enough challenge in learning a concept through spacing and variation so that you avoid entering an ineffective learning “autopilot”, but not so much challenge that you enter academic or personal burnout every time you try to learn something new.

This level of learning difficulty is considered desirable because while it might not appear as successful in the short term, it is shown to greatly improve your ability to learn, retain, and apply information long-term. This long-term retention and confidence in content long-term is much more helpful in skills you are learning to reach a greater academic, personal, or career goal.

I resonated a lot with this podcast episode. I definitely experienced learning “autopilot” in my previous major. I either encountered courses that felt so intuitive that I switched to an autopilot learning mode and didn’t end up retaining anything long term once the class ended. I also didn’t feel motivated to use my understanding of course concepts to ask for more challenging content from my professor.

I have also experienced the opposite of this, I have taken classes that felt initially too intimidating or hard and it dissuaded me entirely from engaging in the class and asking for help when I needed.

Since transferring to DSML, even in this first week, I found myself unknowingly using some of the learning tactics mentioned in the podcast like the study to teach practice by talking through my assignments and lecture notes with my partner who is in an entirely different field.

I also have noticed how the learning methods described in this podcast (variation, pacing, pretesting, and study-to-teach) have been applied in each **class** lecture so far. I have found these methods incredibly helpful in not only retaining my interest in class, but also in motivating me to complete outside-of-class work and review.

This first week has felt challenging, but in a way that is exciting and motivating instead of intimidating. Knowing many of my classmates feel similarly also makes me feel better about engaging in this class.

Keeping this toolbelt of learning skills in mind throughout the rest of my degree will help me be able to naturally practice it in my **post-grad endeavors**. I currently plan to apply to a MPH (masters in public health) program and work in some form of data analysis in the medical field post-graduation, so I know that will come with its own set of educational challenges. Feeling confident in how I learn will definitely make the transition to grad school and then industry less intimidating.

2 Read [Bluesky CEO Jay Graber warns: If you're a student, using AI means](#)

where Bluesky CEO's Jay Graber warns that over-reliance on AI can impair critical thinking and long-term skill development and may contribute to "academic obsolescence" - declining ability to perform without AI

What skills do you risk losing if you depend on AI too much?

As mentioned in the article, skills you risk losing by depending too heavily on AI includes:

- Reduced ability to think critically.
- Reduced ability to problem solve effectively.
- Inability to develop a unique voice and sense of understanding.

"You need to have the good judgment of how you're going to use it, and then you have to have the flexibility to take that knowledge and do something useful with it" (Jay Garber, [timesofindia.com](#)). By utilizing AI as a way to solve problems or find answers each time you encounter a challenge academically, you risk entering a space of "academic autopilot" as mentioned in the podcast assigned earlier in this document. You also lose the ability to defend your stance or answer to a topic because you did not fully engage or comprehend the information provided. An overreliance of AI in academics also leads to further academic disengagement because when you never regularly work through confusion or challenge, you do not develop the skill of identifying what exactly about a subject or topic is challenging.

Instead, AI should be more focused on being utilized as a way to **lightly** guide or direct one to additional resources or as a very brief overview of a topic before then engaging yourself in finding additional content to confirm and affirm your learning takeaways.

Personally, I prefer to avoid using AI tools as much as I can because of my concerns around climate impact and discrimination involved in the location of data centers and the stances of many AI-focused companies. I also refrain from using AI because I know that I do not have the skillset currently to tailor helpful prompts to be able to use AI efficiently in my educational experience. I know as I continue this program, my thoughts and stances may change, and I will also learn and get to practice utilizing AI in Ethical manners academically. But for now, I typically go to any other resource to review content and receive practice sets of questions.

3. Summarize 3 Strategies you will use this semester to build skills, become a lifelong learner, and [Use AI Tools](#) in a way that supports your own thinking.

3 strategies I plan to use this semester include:

- Utilizing AI in personal and academically ethical ways.

For now, I want to refrain from going to AI as a resource and do my best to utilize the resources provided to me (attending lectures, working with classmates, asking labmates, attending office hours) so that I do not risk

isolating myself when I learn and work on assignments. I believe the only things I would ask at the moment would be specific error messages that might not be easily found through an online search.

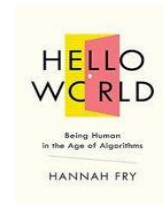
- Prioritizing a desirable difficulties approach to learning

Prioritizing long-term comprehension over short term success in this class will set me up to feel prepared for CS2 as well as any future academic or career endeavors. Practicing methods like variation, pacing, pretesting, and learn-to-teach are ways I can strengthen my personal academic confidence as well as my general skillset throughout the semester.

- Checking in on my collaboration skills

Regularly reviewing my collaboration skills, especially when team work becomes more common in class, will help remind me that the payoff in improving these skills is multifaceted. By understanding my strengths and weaknesses and being upfront about them, I can help promote a positive and nonjudgemental team work environment.

0.3 Algorithms



"You can't just build an algorithm, put it on a shelf and decide whether you think it's good or bad completely in isolation. You have to think about how that algorithm actually integrates with the world that you're embedding in."
Hannah Fry

Read [Chapter 1](#) from Hannah Fry, "Hello World, Being Human in the Age of Algorithms" and answer the following.

Answer the following in your own words.

1. Describe an algorithm.

"...The invisible pieces of code that form the gears and cogs of the modern machine age, algorithms have given the world everything... They have learned our likes and dislikes...And all the while, they have the hidden power to slowly and subtly change the rules about what it means to be human."
(Fry, "introduction")

From the reading, my understanding of an algorithm is a set of commands, typically in a formula, followed to achieve a goal. Algorithms can vary in complexity.

2. Summarize the four main categories of real world tasks that algorithms are doing.

- **Prioritization:** making an ordered list

Ex- Google, which ranks results (prioritization algorithm).

- **Classification:** picking a category
EX- targeted ads (classification algorithm).
- **Association:** finding links
EX-dating websites, Amazon product recommendations
- **Filtering:** isolating what's important
EX- Siri, Alexa.

3. Explain the two main approaches taken by algorithms. Include the pros and cons.

The two main approaches used by algorithms include:

- Rule-based algorithms

EX- instructions for baking a cake.

PROS: rule-based algorithms are typically easy to comprehend/straightforward. They are written logically for any person to follow.
CONS: Only problems where instructions can be written for them can be solved using rule-based algorithms.

- Machine-learning algorithms

EX-teaching a dog a trick.

PROS: Machine-learning algorithms do a better job at solving a problem for which a set of straightforward instructions cannot be applied. I
CONS: The set of steps a machine learning algorithm might take to solve a problem can seem nonsensical, overdone, or unnecessary to the average user. A machine learning-algorithm also runs the risk of misinterpreting a problem or solution entirely, providing a wildly incorrect result.

4. What can happen when we only rely on machines to make decisions for us?

Relying too heavily on a machine in regards to decision making dilutes our ability to critically think and analyze day-to-day situations. It is important to recognize bias's place in technology, and remind ourselves that behind every piece of technology is a set of morals and experiences that fueled its specific development and implementation. Understanding the background of a tool and its potential shortcomings produces a more subjective and critical technology user.

Part 1: Technical Document Explore and Explain

Purpose: Professional communication to demonstrate your learnings by connecting different ideas from lectures or provided resources.

- Clearly explain key ideas in your own words and include supporting evidence such as code examples, images, screenshots, or charts.

You will update your technical document to have it as a resource for your projects, quizzes, future reference and resume.

- [Tech Doc CS1050](#): Download as word or create a copy of the google documentation.

- Put in your git repository after you set up git and github.
- Submit this as a word or pdf with your GE submission.
- Use headings to organize topics and create or update a table of contents to be able to quickly access information.
 - [Headings, Subheadings, and Table of Contents \(Google Docs\)](#)
 - [Headings, Subheadings, and Table of Contents \(Microsoft Word\)](#)
- Complete these sections
 - [Module 1: Programming Fundamentals and Tools](#) all objectives
 - [Version Control](#)
 - [Implement Quality Documented Code](#)
 - Apply Java naming conventions for variables, constants, methods, and classes.
 - Write comments at top of file and inline comments to document code.

Part 2 Analyze and Apply

Demonstrates a clear analysis of a problem and applies relevant concepts effectively. Work shows independent thinking and understanding. You will use Git, GitHub and Eclipse to edit, version and back up your code.

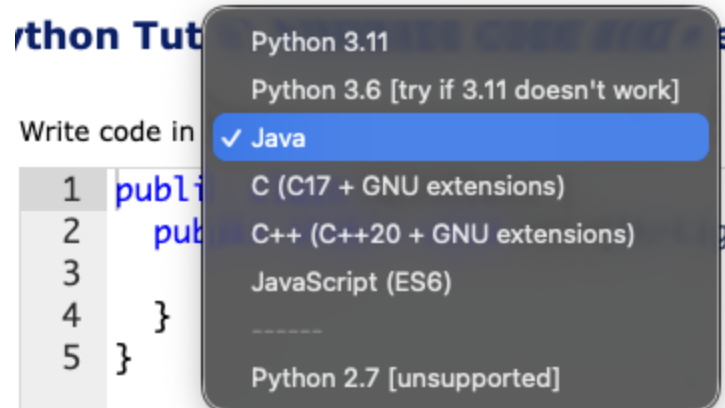
Resources to get started with Git/GitHub and Eclipse

- Git and Github Version Control
- [Setup Eclipse](#) - Download and put in your github Resources Folder
- [Eclipse Cheat Sheet](#) - Download and put in your github Resources Folder

2.1 Explore GuessNumber Code

Be curious about the code as you explore. Come to office hours if you need help setting up your environment to use Eclipse and Git/Github.

1. Complete [Interactive Animation for Guess Number Program](#)
2. Open Eclipse Cheat Sheet and follow the instructions on how to create a project called CS1050M01. Take your time because if you do it wrong it won't work and you will need to do it over.
3. Download [GuessNumber.java](#)
4. Open Eclipse Cheat Sheet and read how to add an existing Java file by dragging and dropping into your CS1050M01 Eclipse project.
 - a. Use [Visualize code in Python, JavaScript, C, C++, and Java](#) if you do not have your environment set up. Select java and paste code so you can edit, run, add comments and step through code.



5. Add comments to the parts of the code to help you understand. You can use parts of the code and comments in your technical document.
6. Version your guess number and technical document locally and then back up on your github cloud server.
7. Upload the .java file into GE M01 submission.

1. Take a screenshot showing you are using the debugger to show what is stored in the variable number and guess after you enter a guess.
2. Describe 3 things you understand better about the code after using the debugger. • • •
3. Identify the warning in the code. Show a screenshot of the warning in the program. Find a resource to help fix the problem based on the warning message. You can use AI for help with syntax errors and warnings.
4. Take a screenshot to show your GuessNumber.java file in your repo on the Github Cloud Server. Explain why only the java file is showing in the repos and not the class file.

2.2 Calculate Grades

You will create a project to store your assignment code and then create your program file. The program you create here will be updated in future guided explorations.

Remember: Commit and push your code often.

1. Use the [Eclipse Cheat Sheet](#) to create a new class called **GEM01GradeCalculations** in your CS1050M01 project
2. Update the code to add an initial comment. **This initial comment is required on all assignments.**
 - a. Your name
 - b. Class name
 - c. Section (M/W) or (T/R) Due date
 - d. Description

```

/*
 * Name: [YOUR NAME]
 * Class: CS1050 (M/W or T/TH)
 * Description: Guided Exploration 01
 * The program will calculate a final grade for this class based on the category weights
 */

public class LastNameFirstNameGE01Calculation
{
    {

    }

    } //main
} //Class GE01

```

3. Design your algorithm on paper. Use an example test case to use to determine how to compute a grade on paper.
 - a. Use constants for percentage weights and variables for your category grades and final grade. Do not hardcode values in your expression to calculate your final grade. Example of hardcoding $(.15 * 95) + (.2 * 92) + (.25 * 88) + (.2 * 87) + (.2 * 93)$
 - b. Use correct naming conventions.

Category Weights	Example Category Grades
Class participation 15%	95
Guided Explorations 20%	92
Quizzes 25%	88
Projects 20%	87
Final Demonstration 20%	93

4. **Working with Variables and Expressions:** Add code to calculate your final grade.

- a. Write a little code and run. Do not try and write all the code at once as this is harder to fix errors.
5. Display your name and final grade using **System.out.println**
6. Add comments to explain your code.
7. Commit your .java code file locally and push your code to github remote repository. Make sure you include comments.
8. Upload the .java file into GE M01 submission.

3 Reflection

1. Identify one specific challenge you faced in this exploration and what you did to overcome it.

2. Team Collaboration

What is one strength of yours when working with your group in class?

What is one strength of your team?

What is one area you would like to improve about your team working together in class?

3. Environment Set Up

On a scale of 1 to 5 rate yourself on using the following tools and include one thing you understand and one thing you need help on.

git/github rating:

- I understand how to
- I need help on

If you are not using Eclipse include what IDE you are using

Eclipse rating:

- I understand how to
- I need help on

4. Now that you are submitting your first Guided Exploration

- What do you wish you would have asked for help on or what did you get help on?
- How was your time management? Did you have to do a lot the day it was due?
- What will you keep the same and what will you try to improve on when you do the next GE?

